# Scoryst

Music engraving in Typst

github.com/bernsteining/scoryst · typst.app/universe/package/scoryst

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# Scoryst — Music Engraving Plugin for Typst

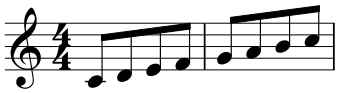
Scoryst is a music notation engraving plugin for Typst. It wraps Verovio as a WebAssembly module, rendering music from multiple input formats directly into SVG embedded in your document.

## Quick Start

Render inline ABC notation:

typst

```typst
#import "../pkg/scoryst.typ": render-music
#render-music("X:1\nM:4/4\nK:C\nCDEF|GABc|")
```



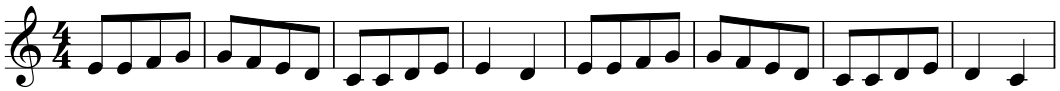
Or define a show rule to render ABC code blocks automatically:

typst

```typst
#import "../pkg/scoryst.typ": render-music
#show raw.where(lang: "abc"): it => render-music(it.text)

```abc
X:1
T:Ode to Joy
M:4/4
K:C
EEFG|GFED|CCDE|E2D2|
EEFG|GFED|CCDE|D2C2|
```
```

Ode to Joy



## API Reference

### `render-music`

typst

```typst
#render-music(
  data,             // string: music data (ABC, MusicXML, MEI, Humdrum, Volpiano, CMME)
  options: none,    // dictionary: verovio options
  page: 1,          // int: page number to render
  ..args,           // forwarded to Typst's image() (width, height, fit, alt)
)
```

### `music-page-count`

typst

```typst
#let n = music-page-count(data, options: none)
```

Returns the number of pages for the given music data and options. Useful to loop over pages of a multi-page score:

## Verovio Options

Options are passed as a Typst dictionary. They map directly to Verovio's toolkit options.

| | | |
|---|---|---|
| **Common** | | |
| `adjustPageHeight` | `true` | Crop SVG height to content |
| `adjustPageWidth` | `false` | Crop SVG width to content |
| `scale` | `100` | Scale factor (percent) |
| `font` | `"Leipzig"` | Music font: Leipzig, Bravura, Gootville, Leland, Petaluma |
| `inputFrom` | `"auto"` | Format: auto, mei, musicxml, abc, humdrum, volpiano, cmme |
| `pageWidth` | `2100` | Page width (MEI units) |
| `pageHeight` | `2970` | Page height (MEI units) |
| `pageMarginTop` | `50` | Top margin |
| `pageMarginBottom` | `50` | Bottom margin |
| `pageMarginLeft` | `50` | Left margin |
| `pageMarginRight` | `50` | Right margin |
| `landscape` | `false` | Landscape orientation |
| `breaks` | `"auto"` | Line breaks: auto, line, encoded, none |
| `condense` | `"auto"` | Condense: auto, none, encoded |
| `transpose` | `""` | Transpose (e.g. "M2" for major second up) |
| `header` | `"auto"` | Header: auto, none, encoded |
| `footer` | `"auto"` | Footer: auto, none, encoded |
| **Layout** | | |
| `spacingStaff` | `12` | Spacing between staves |
| `spacingSystem` | `12` | Spacing between systems |
| `spacingLinear` | `0.25` | Linear spacing factor |
| `spacingNonLinear` | `0.6` | Non-linear spacing factor |
| `unit` | `9` | Base unit size (half staff space) |
| `stemWidth` | `0.2` | Stem width |
| `barLineWidth` | `0.3` | Bar line width |
| `staffLineWidth` | `0.15` | Staff line width |
| `lyricSize` | `4.5` | Lyrics font size |
| `hairpinSize` | `3.0` | Hairpin height |
| **SVG Output** | | |
| `svgViewBox` | `false` | Use viewBox instead of width/height |
| `svgRemoveXlink` | `false` | Use href instead of xlink:href |
| `svgBoundingBoxes` | `false` | Add bounding box rects (debug) |
| `removeIds` | `false` | Strip element IDs from SVG |
| `smuflTextFont` | `"embedded"` | SMuFL text font: embedded, linked, none |

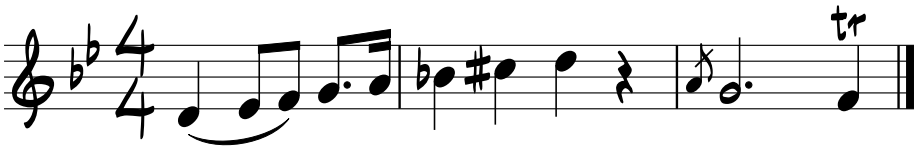
Full reference: https://book.verovio.org/toolkit-reference/toolkit-options.html

## Music Fonts

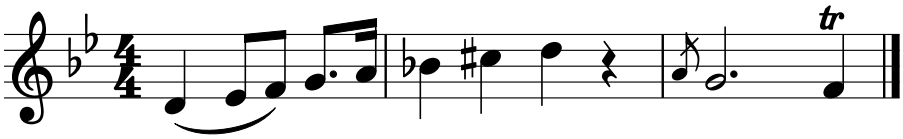
Five SMuFL-compliant music fonts are available. Set the font with the `font` option:

```typst
#render-music(data, options: (font: "Petaluma"))
```

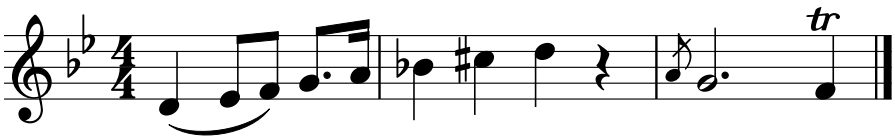
**Petaluma**



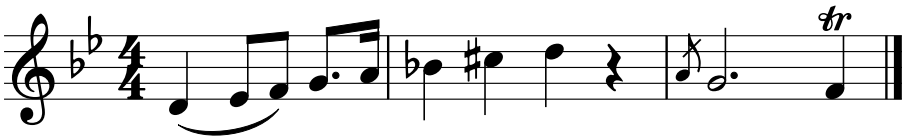
**Leland**



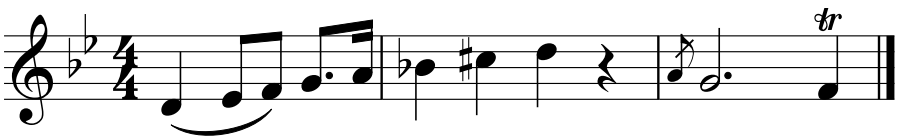
**Gootville**



**Bravura**



**Leipzig**



## Supported Input Formats

Verovio auto-detects the input format for ABC, MusicXML, MEI, and Humdrum. For Volpiano and CMME, pass `inputFrom` explicitly.

All the files used in the examples are available in the project's Github.

### Exporting from notation software

If you're working with any of these scoring software this table sums up each supported export format with their documentation linked.

| Software | MusicXML | MEI | ABC | Humdrum |
|---|---|---|---|---|
| MuseScore | ✓ | ✓ | | |
| Finale | ✓ | | | |
| Sibelius | ✓ | | | |
| Dorico | ✓ | | | |
| LilyPond | ✓ | | | |
| Flat.io | ✓ | | | |
| Noteflight | ✓ | | | |
| ABC tools | ✓ | | ✓ | |
| Humdrum tools | ✓ | ✓ | ✓ | ✓ |
| MEI tools | ✓ | ✓ | | |

MusicXML is the universal interchange format — virtually all notation software can export it. For the best results, export as uncompressed MusicXML (`.musicxml` or `.xml`, not `.mxl`).

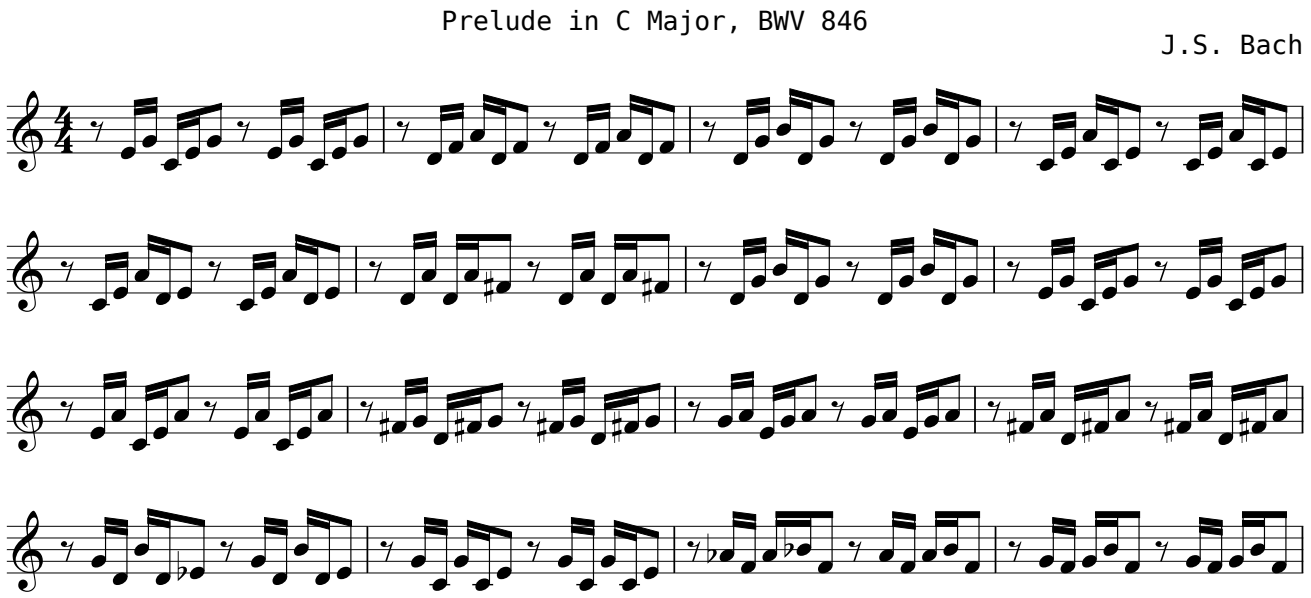
## ABC Notation

ABC specification · Download ABC files

ABC is a compact text-based format popular for folk and classical melodies. Chord symbols are supported.

typst

```
#render-music(read("bach-prelude-cmaj.abc"))
```



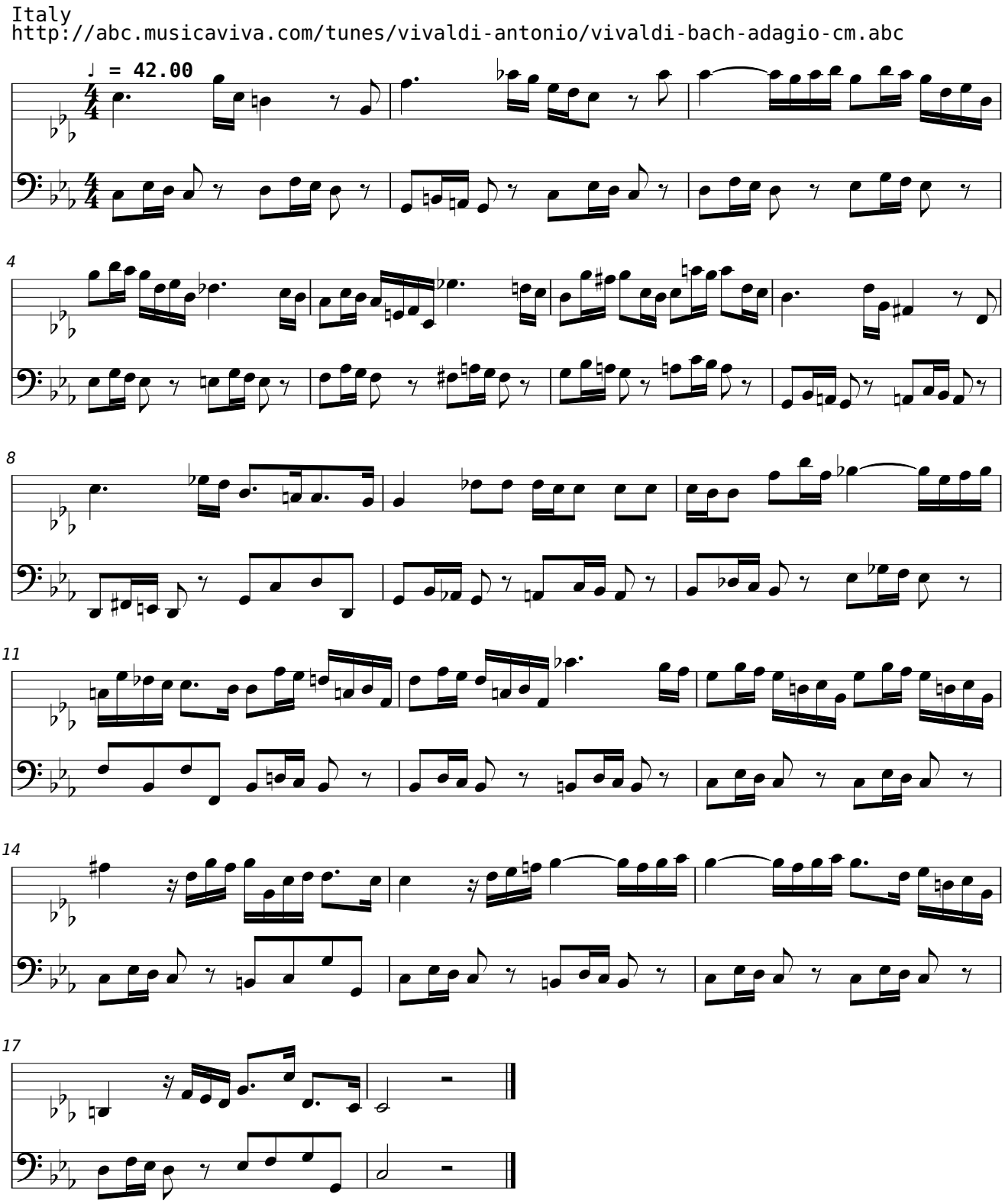
## MusicXML

MusicXML 4.0 specification · Download MusicXML files

MusicXML is the standard interchange format for notation software. It supports grand staff, multiple voices, dynamics, and full score layout.

typst

```typst
#render-music(read("adagio.xml"))
```



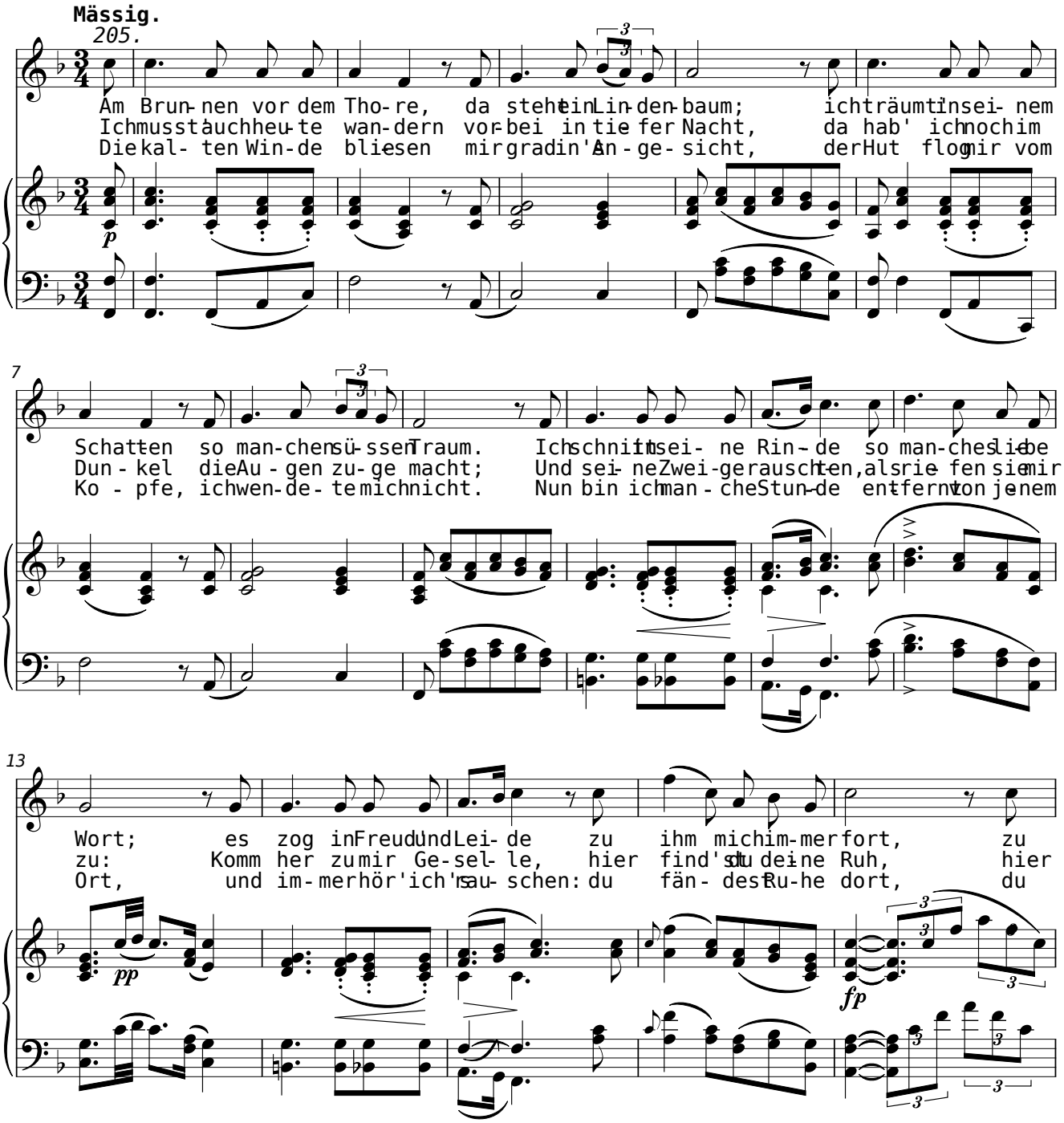
## MEI (Music Encoding Initiative)

MEI Guidelines · Sample MEI encodings

MEI is a rich XML-based format used in musicology, supporting lyrics, polyphonic textures, fermatas, and detailed editorial markup.

typst

```
#render-music(read("schubert.mei"))
```



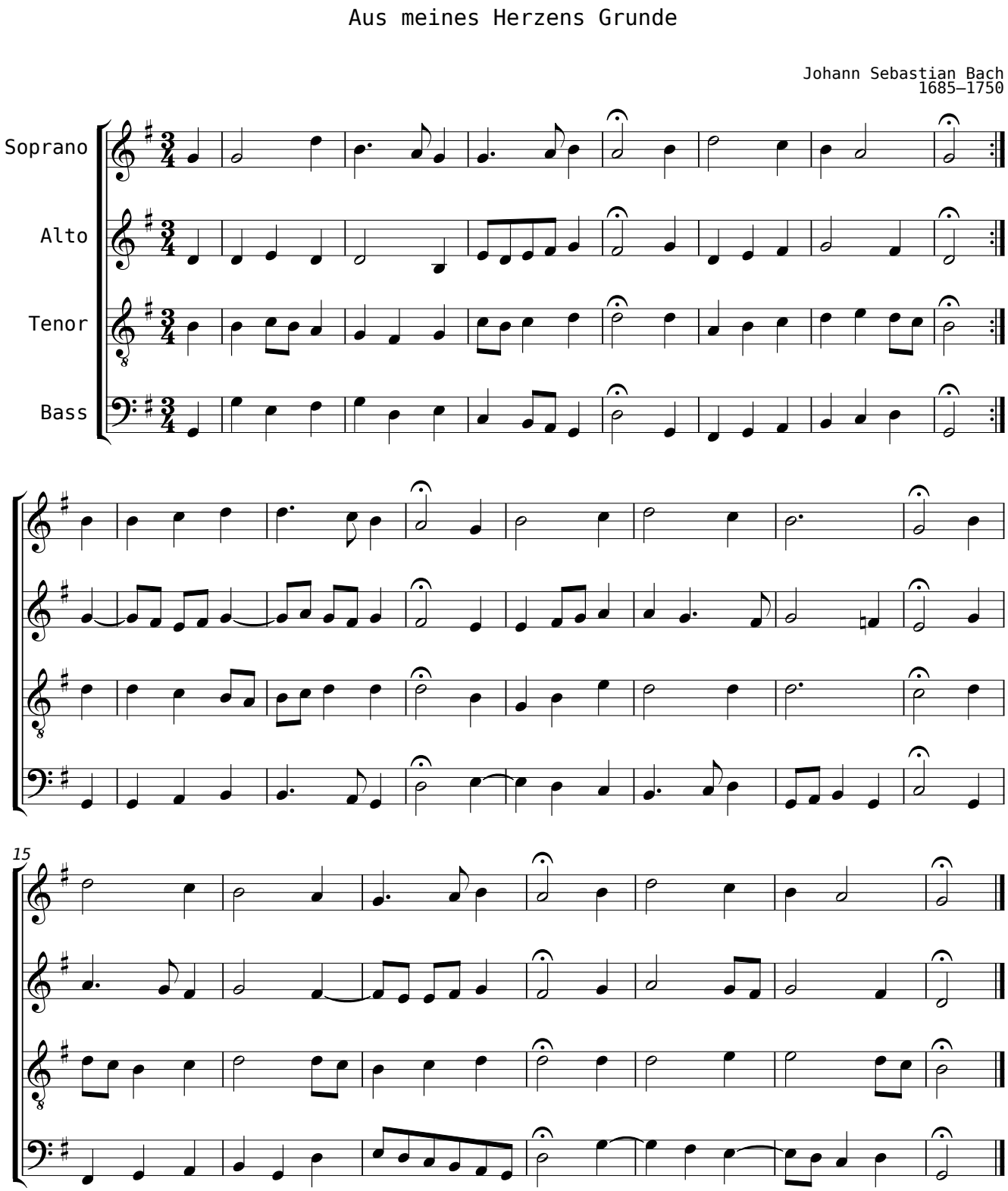
## Humdrum

Humdrum User Guide · Download kern files

Humdrum uses a tab-separated spine structure with **kern encoding for pitches and durations. Widely used in computational musicology.

typst

```
#render-music(read("sample-humdrum.krn"))
```



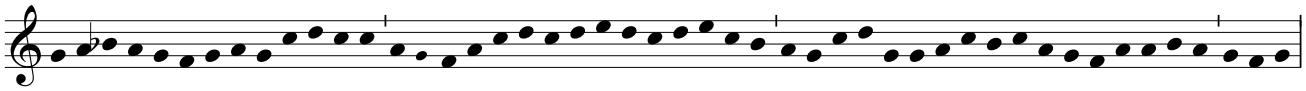
## Volpiano

Volpiano specification · CANTUS database

Volpiano is a text encoding for medieval chant notation, used by the CANTUS database. Here is *Veni Creator Spiritus*, the famous Pentecost hymn. Requires `inputFrom: "volpiano"`.

typst

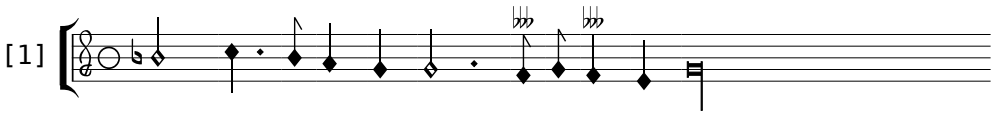
```
#render-music("1---g--hij---hgf--g--hg---k--lk--k7---hG--f---h--k--lk---l--m--l---k--lm---kj7--hg--kl---g--gh--k---
jk---h---gf--h--hjh7---g--f--g7---3", options: (inputFrom: "volpiano"))
```



## CMME

CMME project · Download CMME files

CMME is an XML format for mensural notation (medieval and Renaissance music). Requires `inputFrom: "cmme"`.

typst

```
#render-music(read("cmme.xml"), options: (inputFrom: "cmme"))
```